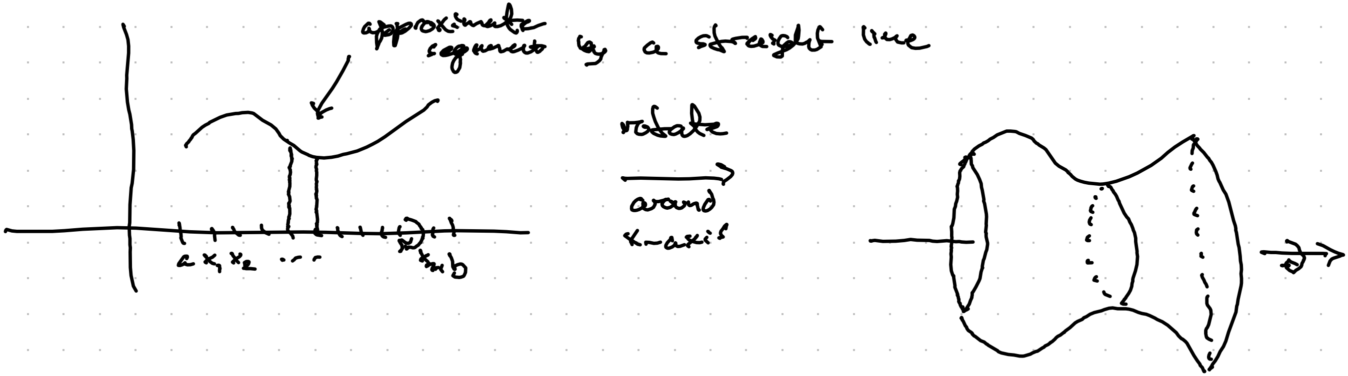
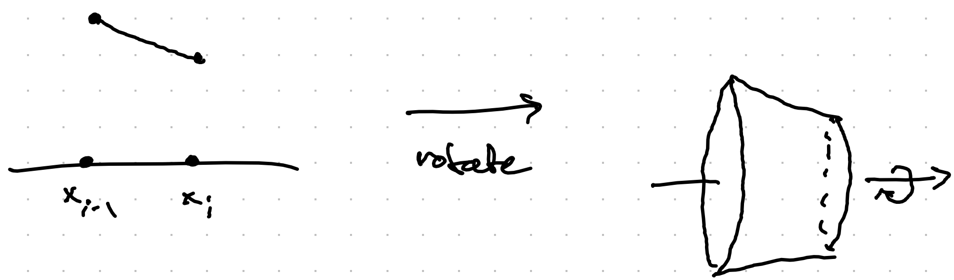
The surface area formula – 2.4



<u>Calc I</u>: wanted to find the <u>volume</u> of such "solids of rotation" obtained by rotating <u>area</u> around an axis.

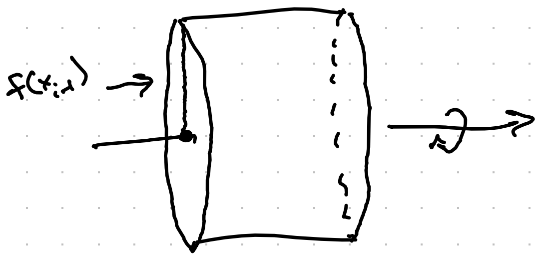
<u>Calc II</u>: want now to find the <u>area</u> of these "surfaces of rotation"

Zoom in on the interval $[x_{i-1}, x_i]$:



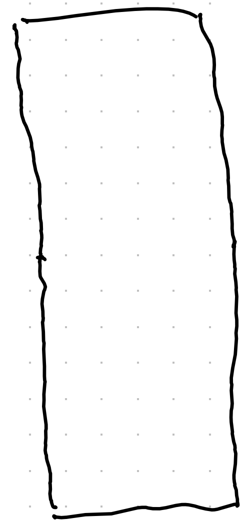
Goal: find the surface area of these pieces and then add them all up.

Nice special case: $f(x_{i-1}) = f(x_i)$ and our straight-line approximation to the curve is completely horizontal.



Rotation around x-axis produces a cylinder!

Cutting and unfolding gives a rectangle:

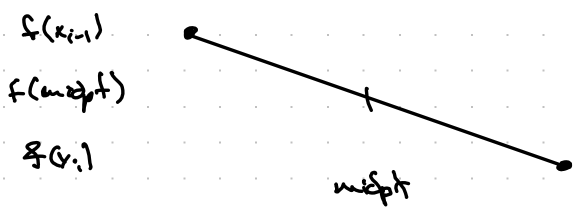


height is the circumference $2\pi r$ $= 2\pi f(x_i)$

base $= x_i - x_{i-1}$

Surface area in this case is $2\pi f(x_i)(x_i - x_{i-1})$.

When $f(x_{i-1}) \neq f(x_i)$, find the average circumference $2\pi \cdot \underbrace{\frac{f(x_{i-1}) + f(x_i)}{2}}_{f(\text{midpt of interval})} = \pi\left(f(x_{i-1}) + f(x_i)\right)$



Avg circumference is

$$2\pi f(\text{midpt})$$

Surface area contribution from $[x_{i-1}, x_i]$ is

$$\underbrace{2\pi f(\text{midpt})}_{\substack{\text{average} \\ \text{circumference}}} \underbrace{\sqrt{1 + f'(x_i^*)^2}\,\Delta x}_{\substack{\text{curve segment} \\ \text{length}}} \xrightarrow[\substack{n \text{ and} \\ n \to \infty}]{\substack{\text{sum} \\ \text{over}}} \int_a^b 2\pi f(x)\sqrt{1 + f'(x)^2}\,dx$$